

Wonkyung (Woni) Choi

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RESEARCH INTERESTS	Clinical Neuroengineering, Brain-Computer Interface, Neuromodulation, Deep Brain Stimulation, Eating Disorders, Obsessive-Compulsive Disorder, Compulsive behaviors, Computational and theoretical Neuroscience, Systems Neuroscience, Interoception	
EDUCATION	University of Pennsylvania <i>Ph.D. candidate, Bioengineering</i>	Philadelphia, PA Aug. 2022 - Present
	University of Pennsylvania <i>M.S.E., Bioengineering</i>	Philadelphia, PA Aug. 2025
	Sogang University <i>B.A., Psychology & B.S., Physics Summa Cum Laude</i>	Seoul, South Korea Aug. 2021
	Boston College <i>Exchange Student</i>	Boston, MA Spring 2019
RESEARCH EXPERIENCES	University of Pennsylvania <i>PhD Student (Advisor: Dr. Casey Halpern)</i>	Philadelphia, PA Aug. 2022 - Present
	- Working on a clinical trial management and an electrophysiological data analysis for nucleus accumbens responsive neurostimulation (NAc-RNS) to ameliorate loss of control (LOC) eating (NCT03868670) - Working on an electrophysiological data analysis to investigate biomarkers and stimulation effect on treatment-resistant OCD patients enrolled in the SEEG-guided multi-lead deep brain stimulation clinical trial (NCT05623306)	
	Sogang University <i>Undergraduate Research Assistant (Advisor: Dr. Joon I. Jang)</i>	Seoul, South Korea Jan. 2021 - Nov. 2021
	Co-first authored a paper reporting on the impact of electron-hole exchange interaction on the biexciton kinetics and thermodynamics in pristine and Mn-doped (BA)₂PbBr₄. Worked on precision laser spectroscopy experiments and analysis (MATLAB, Origin).	
PUBLICATIONS	Sungkyunkwan University Center for Neuroscience Imaging Research <i>Summer Intern (Advisor: Dr. Choong-Wan Woo)</i>	Suwon, South Korea Jun. 2018 - Jul. 2018
	Peer-reviewed Journals	[†] equal contribution
	Seilheimer, R.L.; [†] Vardalakis, N.; [†] Choi, W. ; Qiu, L.; Campos, G.; Nho, Y.; Chang, A.; Brown, L.A.; Oathes, D.J.; Ganguly, T.; Cristancho, M.; Barbosa, D.A.N.; Pesaran, B.; Halpern, C.H.; Scangos, K.W.; Personalized Network Biomarkers of Deep Brain Stimulation Response in Obsessive-Compulsive Disorder, under preparation	
	Chang, A.; Michiels, J.; Campos, G.; Nho, Y.; Duprat, R.; Vardalakis, N.; Koluda, A.; Qiu, L.; Choi, W. ; Seilheimer, R.L.; Ganguly, T.; Hamilton, R.; Barbosa, D.A.N.; Scangos, K.W.; Oathes, D.J.; Halpern, C.H.; Vitale, F.; A Reproducible Safety Assessment Model for Concurrent Transcranial Magnetic Stimulation with Implanted Intracranial Electroencephalography Electrodes, under revision	
	Choi, W. , [†] Nho, Y.; [†] Qiu, L.; Chang, A.; Campos, G.; Seilheimer, R.L.; Wilent, W.B.; Bakalov, D.; Firdous, N.; Kerr, M.; Joshi, D.; Maze, G.; Topalovic, U.; Batista, D.; Suthana, N.; Amaro, A.; Hayes, M.R.; Cajigas, I.; Cristancho, M.; Allison, K.C.; Pesaran, B.; Scangos, K.W.; Gold, J.I.; Wadden, T.A.; Halpern, C.H. Brain activity associated with breakthrough food preoccupation in an individual on tirzepatide , <i>Nat. Med.</i> , 2025 , doi:10.1038/s41591-025-04035-5	
	Choi, W. , [†] Nam, S.H.; [†] So, H.K.; Lee, S.E.; Jung, M.H.; Jang, J.I. Impact of Dark Excitons on the Population and Relaxation Kinetics of Two-Dimensional Biexcitons in [CH₃(CH₂)₃NH₃]₂Pb_{1-x}Mn_xBr₄ (x = 0–0.09) , <i>J. Am. Chem. Soc.</i> , 2021 , <i>143</i> , 19785–19793. doi:10.1021/jacs.1c08474	

**ORAL
PRESENTATIONS**

Penn Bioengineering Graduate Research Symposium May. 2025, Philadelphia, PA
“Tirzepatide amelioration of dysregulated eating behavior is associated with human accumbens delta-theta oscillations”

**American Society for Stereotactic and Functional
Neurosurgery Biennial Meeting
Psychiatric Neurosurgery Symposium** Jun. 2024, Nashville, TN
“GLP-1RA amelioration of food preoccupation is associated with human accumbens delta band attenuation in a single subject”

**POSTER
PRESENTATIONS**

10th Annual BRAIN initiative conference Jun. 2024, Rockville, MD
“GLP-1 Receptor Agonist Ameliorates Food Preoccupation and May Suppress Delta Band Signal in the Human Nucleus Accumbens” [*Scholar Spotlight Honorable Mention*]

**American Society for Stereotactic and Functional
Neurosurgery Biennial Meeting** Jun. 2024, Nashville, TN
“GLP-1 Receptor Agonist Ameliorates Food Preoccupation and May Suppress Delta Band Signal in the Human Nucleus Accumbens”

Systems Neuroscience Symposium Jul. 2018, Suwon, South Korea
“Decoding Context: how story changes brain activities”

HONORS

Dean’s list, Sogang University 2015 - 2016
• Top 1% in the Social Sciences Department

Honors Scholarships, Sogang University Fall 2015, Spring 2017 - Fall 2017
• Merit-based scholarships provided to top ~15% of students in the Social Sciences Department (by GPA) per semester

CORE SKILLS

- Computational tools: MATLAB, C, Python, OriginLab
- Experienced systems: NeuroPace, scalp EEG (brainvision), BCI2000, g.tec
- Experimental skills: Precision solid-state laser spectroscopy
- Languages: Korean (native), Japanese (basic proficiency)

**LEADERSHIP
& SERVICES**

- *Mentor*, Penn BE Applicant Support Program 2024 - 2026
- *Mentor*, Study Abroad Information Seminar, Sogang University 2024 - 2026
- *NGO Delegate from Anglican Church of South Korea*,
UN CSW60 (60th Session of the UN Commission on the Status of Women) Mar. 2016
- *Founder & President*, Ttanjit - Classics Reading Club 2015 - 2016
- sponsored by the Center for Teaching and Learning in Sogang University
- *Member*, H.U.G (Hands Up for Gathering) 2015
- volunteering club for helping exchange students in Sogang University